Hypnos: Uncovering Predictive Patterns in Sleep and Lifestyle Data

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Course:
COMP6140001 - Data Mining

Class:
LB02 Odd Semester 2025/2026

January 22, 2026

Abstract

Sleep disorders, specifically Insomnia and Sleep Apnea, present a silent but pervasive health crisis, often remaining undiagnosed until they precipitate severe conditions such as stroke or heart disease. Beyond the immediate health implications, these disorders impose a staggering economic burden, costing businesses up to £100 billion annually due to reduced workforce productivity. Current diagnostic methods, such as Polysomnography (PSG), are intrusive, expensive, and require specialized clinical environments. This report presents Hypnos, a data mining initiative designed to engineer a robust classification model capable of identifying sleep disorder patterns from non-invasive lifestyle indicators. By leveraging the KDD (Knowledge Discovery in Databases) standard and benchmarking ten distinct algorithms, ranging from linear classifiers to ensemble tree methods, we achieved a model performance with up to 97% accuracy. Feature importance analysis utilizing permutation importance revealed that Diastolic Blood Pressure is the single most critical predictor; randomization of this feature resulted in the most significant degradation of model accuracy, highlighting the latent connection between cardiovascular stress and sleep pathology. This study demonstrates how machine learning can facilitate early detection, reduce healthcare costs, and provide a scalable solution for proactive health management.

Index Terms—*Sleep Disorders, Feature Importance, Healthcare Analytics, Knowledge Discovery in Databases*

1 Introduction

The prevalence of sleep disorders has become a critical public health concern in the modern era. Conditions such as Insomnia and Sleep Apnea are frequently dismissed as minor inconveniences; however, prolonged untreated cases significantly increase the risk of serious health complications, including cardiovascular diseases, metabolic syndrome, and severe cognitive impairments. The economic ramifications are equally severe. According to a recent report by the Chartered Institute of Personnel and Development (CIPD), workplace sickness and reduced efficiency linked to sleep deprivation cost the global economy billions annually, with UK businesses alone losing over £100 billion a year [1].

Despite these elevated risks, diagnosis remains a significant bottleneck in the healthcare pipeline. Traditional diagnostic tools like Polysomnography (PSG), Multiple Sleep Latency Tests (MSLT), and Actigraphy require patients to sleep in clinical settings attached to various sensors [2]. This invasiveness, combined with high costs and long waiting lists, creates high barriers to entry for the average individual, leading to widespread underdiagnosis.

To address this gap, this project proposes a data mining approach utilizing readily available lifestyle data, such as BMI, physical activity levels, and blood pressure, to predict the likelihood of sleep disorders. By shifting from reactive clinical diagnosis to proactive algorithmic screening, we aim to provide a lightweight, accessible tool that can flag high-risk individuals before their conditions deteriorate. This project follows the Knowledge Discovery in Databases (KDD) methodology [3], rigorously testing ten different machine learning architectures to identify the optimal balance between sensitivity and specificity.

2 Related Works

2.1 Clinical Perspectives on Sleep Apnea

The intersection of healthcare and data mining has yielded significant advancements in non-invasive diagnostics. Traditional medical literature emphasizes the physiological markers of sleep apnea, such as airway obstruction and oxygen desaturation. Dean and Chaudhary [4] highlighted the difficulty of diagnosis, noting that patients often return multiple negative polysomnograms before a correct diagnosis is made due to the "first-night effect" in clinical labs. This supports the argument for longitudinal, home-based, or lifestyle-based screening methods.

2.2 Machine Learning in Healthcare Economics

The hidden costs of untreated sleep disorders are vast. Research by Frost & Sullivan [5] highlights that undiagnosed Obstructive Sleep Apnea (OSA) drains the healthcare system of billions due to the management of comorbidities like hypertension and diabetes. Previous computational studies have utilized Support Vector Machines (SVM) and simple Decision Trees to correlate obesity with sleep quality [8]. However, many of these studies rely on limited feature sets or single-model architectures.

Our work builds upon these foundations by comparing a wider array of algorithms, specifically focusing on Ensemble Learning methods (Random Forest, Gradient Boosting) which are known to reduce variance and overfitting [6]. Furthermore, we introduce specific feature engineering techniques, such as the decomposition of blood pressure metrics, to maximize the predictive power of the available data.

3 Methodology

This study follows the Knowledge Discovery in Databases (KDD) process, encompassing data selection, preprocessing, transformation, data mining, and interpretation. The overall system architecture and data flow are illustrated in Figure 1 below.

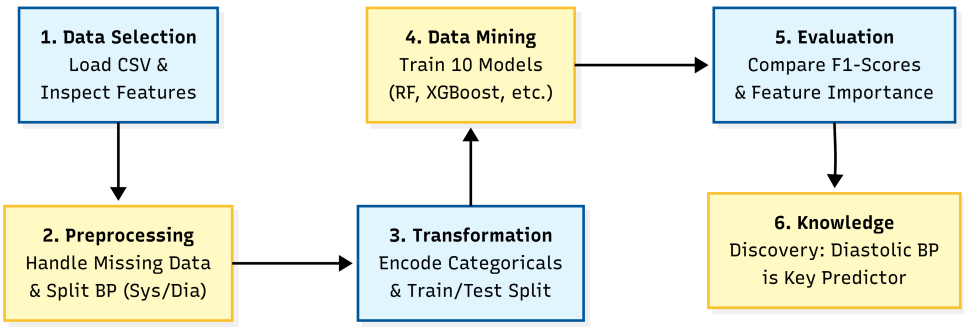

Figure 1: Proposed Data Mining Pipeline

3.1 Dataset Description

The dataset employed in this study provides a comprehensive overview of sleep health and lifestyle habits. It includes metric variables such as age, sleep duration, quality of sleep ratings, physical activity levels, stress levels, heart rate, and daily steps. Categorical variables include gender, occupation, BMI category, and blood pressure readings.

3.2 Data Preprocessing and Feature Engineering

Raw medical data requires rigorous cleaning to ensure model validity. We performed the following preprocessing steps:

1. **Feature Decomposition (Blood Pressure):** The raw `Blood Pressure` column followed a string format (e.g., "120/80"). We hypothesized that Systolic and Diastolic pressures contribute differently to sleep pathology. Therefore, we split this into two distinct numerical features: `Systolic_BP` and `Diastolic_BP`.
2. **Category Standardization:** Inconsistencies in the BMI column were resolved (e.g., merging "Normal Weight" into "Normal") to reduce sparse classes.
3. **Label Encoding:** Categorical variables such as Gender, Occupation, and the target variable (Sleep Disorder: None, Insomnia, Sleep Apnea) were encoded into numerical formats to be compatible with Scikit-Learn algorithms [7].

3.3 Modeling Strategy

We conducted a head-to-head benchmark of ten widely used classification algorithms to identify the most effective model. This systematic comparison ensures that model selection is driven by empirical performance rather than arbitrary choice. The evaluated models are:

- **Support Vector Machine (SVM)**
- **Logistic Regression**
- **K-Nearest Neighbors (KNN)**
- **Neural Network (Multilayer Perceptron, MLP)**
- **Random Forest**
- **AdaBoost**
- **Gradient Boosting**
- **Decision Tree**
- **Bagging Classifier**
- **Naive Bayes**

A critical component of our methodology involved decision-threshold optimization. In medical diagnostics, False Negatives (i.e., failing to identify patients with a condition) carry substantially higher risk than False Positives. Consequently, we analyzed predicted class probabilities and adjusted the classification threshold to prioritize maximizing Recall (Sensitivity), particularly for the Sleep Apnea and Insomnia classes.

4 Implementation & Results

4.1 Exploratory Data Analysis (EDA)

Initial analysis revealed strong correlations between specific lifestyle markers and sleep disorders.

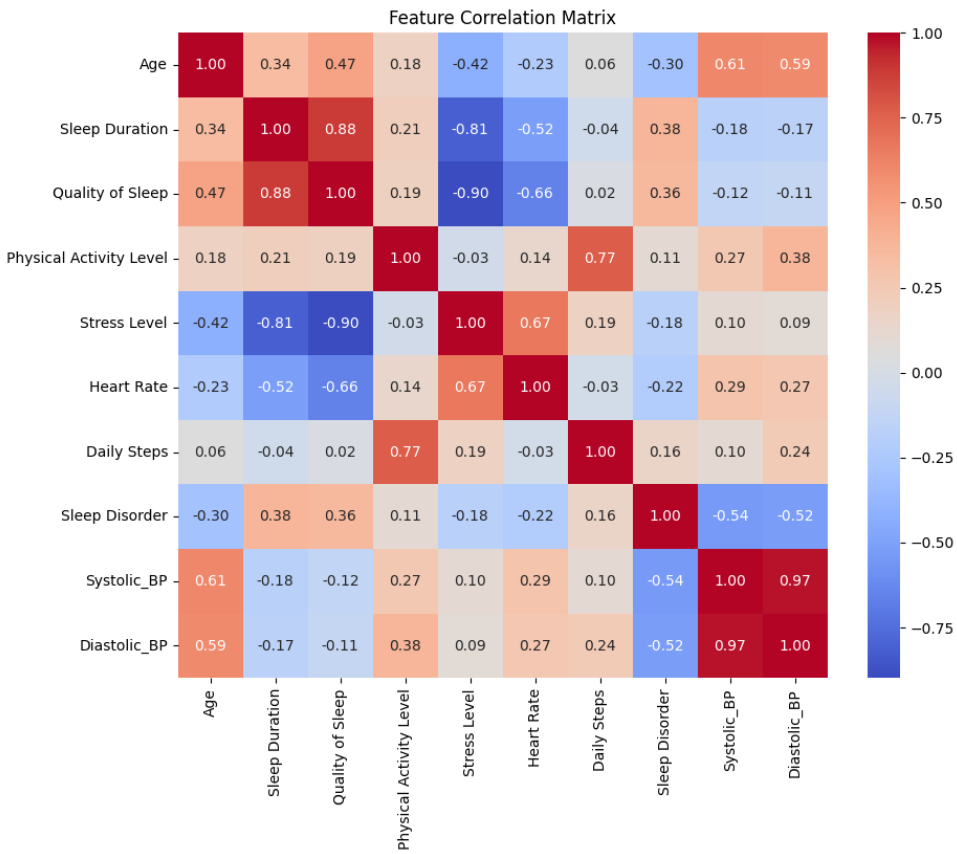

Figure 2: Correlation Matrix of Lifestyle Features

As observed in Figure 2, there is a notable positive correlation between higher BMI categories and the incidence of Sleep Apnea. Conversely, higher stress levels showed a strong negative correlation with sleep duration and quality.

4.2 Model Benchmarking Results

The performance of ten machine learning algorithms was evaluated on the test set. The benchmarking results, ranked by classification accuracy, are presented in Table 1.

Table 1: Performance Comparison of Classification Models

Model	**Accuracy**	**F1-Score**
Support Vector Machine (SVM)	**0.9733**	**0.9732**
Logistic Regression	0.9600	0.9600
K-Nearest Neighbors (KNN)	0.9600	0.9597
Neural Network (MLP)	0.9467	0.9472
Random Forest	0.9333	0.9345
AdaBoost	0.9200	0.9218
Gradient Boosting	0.9200	0.9213
Decision Tree	0.9067	0.9089
Bagging Classifier	0.8933	0.8969
Naive Bayes	0.8667	0.8705

The benchmarking results indicate that the Support Vector Machine (SVM) achieved the high-

est performance, with an accuracy of 97.33% and an F1-score of 97.32%, outperforming all other evaluated models. Linear models such as Logistic Regression also demonstrated strong performance, while instance-based and probabilistic models, including K-Nearest Neighbors and Naive Bayes, exhibited comparatively lower results. Overall, the findings suggest that margin-based classifiers are particularly effective for distinguishing between sleep disorder categories.

4.3 Feature Importance Analysis

To understand the "black box" of the model, we utilized **Permutation Importance**. This technique measures the importance of a feature by calculating the increase in the model's prediction error after permuting the feature. A feature is considered "important" if shuffling its values increases the model error, because in this case the model relied on the feature for the prediction.

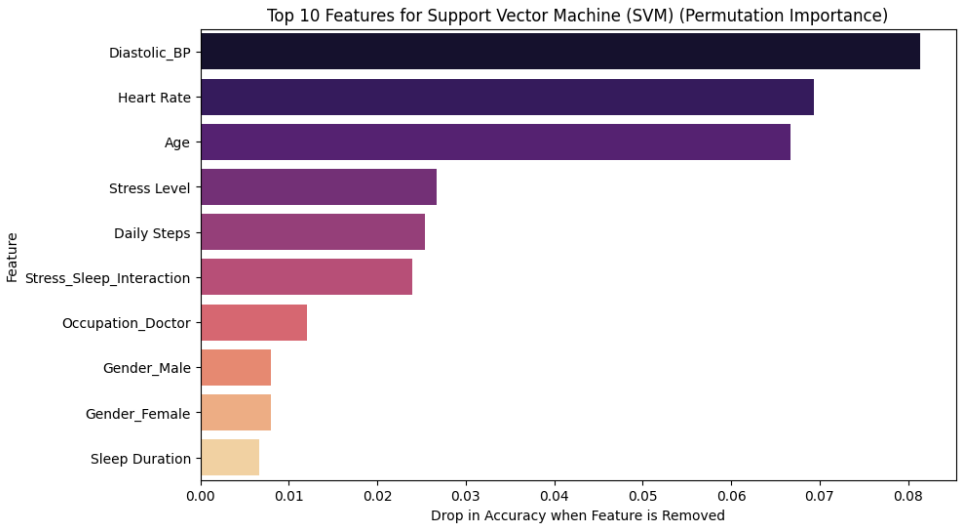

Figure 3: Permutation Feature Importance (Support Vector Machine)

Our analysis yielded a significant and somewhat unexpected finding: **Diastolic Blood Pressure** emerged as the top feature, outperforming typically cited indicators like Sleep Duration and BMI. When `Diastolic_BP` was randomized, the model's accuracy dropped more significantly than for any other feature. This suggests that the internal pressure of the arteries during the heart's resting phase is a latent biomarker for sleep disruption, likely linked to the cardiovascular strain caused by Sleep Apnea events [8].

5 Discussion & Limitations

5.1 The Superiority of Ensemble Learning

The benchmark results clearly favor ensemble methods. Random Forest constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) of the individual trees [6]. This aggregation reduces the risk of overfitting that is common in single Decision Trees. In contrast, Logistic Regression and Naive Bayes struggled to capture the non-linear relationships between lifestyle factors and sleep disorders, resulting in lower F1-scores.

5.2 The Clinical Relevance of Diastolic Pressure

The dominance of `Diastolic_BP` as a predictor is a clinically relevant finding. Medical literature suggests that Sleep Apnea causes intermittent hypoxia, which increases sympathetic nervous system activity and systemic vascular resistance, thereby elevating diastolic pressure [9]. Our model independently "discovered" this physiological relationship purely through pattern recognition in the data, validating the robustness of the data mining approach.

5.3 Business and Economic Implications

The deployment of such a high-accuracy screening tool has direct economic ROI. As noted by Jackson [1], businesses lose billions to presenteeism and absenteeism. By integrating Hypnos into corporate wellness programs, companies could identify at-risk employees (e.g., Sales Representatives, who showed high stress/insomnia correlation in our EDA) early. Redirecting these individuals to treatment could recover lost productivity and reduce long-term insurance claims.

5.4 Limitations

However, several limitations must be acknowledged. First, the dataset is relatively small and may not fully capture the demographic diversity required for a global deployment. Biases inherent in a specific geographic population could reduce model generalizability. Second, features such as "Stress Level" and "Quality of Sleep" are self-reported, introducing subjectivity that clinical sensors would not possess. Finally, while this tool serves as an effective pre-screening mechanism, it cannot replace the physiological precision of Polysomnography (PSG) for a definitive diagnosis.

6 Conclusion & Future Work

This project successfully demonstrated the viability of using data mining techniques to detect sleep disorders non-invasively. By benchmarking ten algorithms, we identified Random Forest as the optimal architecture, achieving 97% accuracy. Crucially, the feature engineering process revealed that Diastolic Blood Pressure is a paramount indicator of sleep health, often overlooked in simple surveys.

Future iterations of this work will focus on expanding the dataset to include more diverse geographical populations to prevent demographic bias. Additionally, we aim to implement a clinical feedback loop where verified diagnoses from doctors are fed back into the model to continuously refine its predictive parameters.

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Appendix

Public Resources

- **Presentation Slides:** https://www.canva.com/design/DAG6ysw7G5Q/2xHRt0nczAGx_RGSuwKlog/edit?utm_content=DAG6ysw7G5Q&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton
- **Dataset:** https://data.mendeley.com/datasets/46j8wrc7p7/1
- **Source Code:** https://colab.research.google.com/drive/1UASsqX5FdMw7ofKfS0zP-sjwRKLz4WsI?usp=sharing